

01

A 60-year-old man presents with shortness of breath and pleuritic chest pain. Chest X-ray reveals a large pleural effusion. Thoracentesis is performed, and the pleural fluid analysis shows: WBC count 15,000/mm³, glucose 80 mg/dL, LDH 800 U/L. Gram stain reveals gram-positive diplococci. Which of the following pleural fluid characteristics is the strongest indication for chest tube drainage?

☐ LDH 800U/L

☐ Glucose 90mg/dL

☒ Gram stain diplococci in chain

☐ WBC 15,000/mm³

Explanation :

Pleural effusions can result from infectious, malignant, or other inflammatory causes. Thoracentesis with pleural fluid analysis helps differentiate between transudative and exudative effusions and guides management.

Infectious effusions are classified into:

- Uncomplicated parapneumonic effusion: Does not require drainage.
- Complicated parapneumonic effusion: Requires drainage to prevent progression to empyema.
- Empyema: Presence of pus in the pleural space or positive Gram stain/culture, requiring urgent chest tube drainage.

Gram stain showing gram-positive diplococci is diagnostic of an infected effusion and indicates empyema, which necessitates immediate drainage via a chest tube to evacuate the infected material. LDH 800 U/L suggests an exudative effusion (LDH > 2/3 upper limit of serum normal per Light's criteria), but it alone does not mandate drainage. WBC 15,000/mm³ indicates inflammation but does not solely determine the need for drainage.

- UpToDate: "Management of parapneumonic effusion and empyema in adults."

02

A 65-year-old man with stage 3 chronic kidney disease (CKD) presents for a routine follow-up appointment. His most recent hemoglobin level is 9.5 g/dL. According to current guidelines, what is the recommended target hemoglobin range for patients with CKD?

☐ 9-10 g/dL

☒ 10-11.5 g/dL

☐ 11.5-13 g/dL

☐ 13-15 g/dL

Explanation :

Anemia is a common complication of CKD due to reduced erythropoietin production. Management involves erythropoiesis-stimulating agents (ESAs) to target an appropriate hemoglobin range. Current recommendations (KDIGO guidelines):

- ESAs should be used to maintain hemoglobin levels above 10 g/dL to reduce the need for transfusions and improve symptoms.
- Target hemoglobin should not exceed 11.5 g/dL, as higher levels are associated with increased cardiovascular risk (e.g., stroke, thromboembolism).
- Targeting 13–15 g/dL is contraindicated because of adverse outcomes, including increased mortality in CKD patients.

The patient's hemoglobin is 9.5 g/dL, below the threshold for initiating ESA therapy. Treatment should aim for 10–11.5 g/dL, consistent with the guidelines, to improve anemia without increasing cardiovascular risk.

- KDIGO 2012 Clinical Practice Guidelines for Anemia in CKD, UpToDate: "Anemia in CKD: Treatment with ESAs."



A public health team plans to conduct active surveillance for tuberculosis (TB) in a specific community for one year. The team will perform door-to-door screening, including symptom assessment, tuberculin skin testing (TST), and chest X-rays as needed. Which type of surveillance is being implemented?

- ☒ Active surveillance
- ☐ Passive surveillance
- ☐ Sentinel surveillance
- ☐ Syndromic surveillance

Explanation :

- Active surveillance: Public health teams actively seek cases through screenings, tests, and interventions. Door-to-door symptom assessment and targeted testing (as in this case) are hallmark examples.
- Passive surveillance: Relies on healthcare providers reporting cases they diagnose.
- Sentinel surveillance: Involves selected institutions or groups monitoring specific diseases to provide data for broader population trends.
- Syndromic surveillance: Uses real-time monitoring of symptoms or health-seeking behaviors (e.g., emergency department visits) to identify outbreaks early.

The described scenario involves door-to-door active case finding, TST, and chest X-ray screening to identify undiagnosed TB cases. This is a textbook example of active surveillance, aimed at reducing TB burden in the community.

- UpToDate: "Epidemiology of tuberculosis."

A 28-year-old man presents with a 6-month history of persistent diarrhea, characterized by frequent, loose stools. He also reports painful ulcerations in his mouth and around his anus. He has experienced significant weight loss and fatigue. Physical examination reveals pale conjunctivae, angular cheilitis, and aphthous ulcers in the oral mucosa. There are also painful, shallow ulcers in the perianal region. Which of the following trace elements is most likely deficient in this patient?

- ☐ Iron
- ☐ Niacin
- ☐ Selenium

☒ Zinc

Explanation :

Persistent diarrhea: Chronic diarrhea causes excessive zinc loss, as zinc is lost in feces during intestinal inflammation. Perianal and oral ulcers: Zinc is critical for epithelial repair and immune function. Deficiency leads to impaired wound healing and epithelial breakdown. Other findings like angular cheilitis, pale conjunctiva (anemia), and significant weight loss are also commonly associated with malabsorption syndromes that cause trace element deficiencies, particularly zinc.

Causes of Zinc Deficiency:

- Malabsorption (e.g., Crohn's disease, celiac disease).
- Chronic diarrhea (increased zinc loss).
- Inflammatory bowel disease, which also explains this patient's systemic symptoms and perianal findings.

Iron would primarily manifest as anemia with fatigue and pale conjunctiva, but it does not explain the perianal and oral ulcers or chronic diarrhea. Niacin deficiency (pellagra) presents as diarrhea, dermatitis, and dementia ("3 Ds"), but it does not cause ulcerations. Selenium deficiency leads to cardiomyopathy or hypothyroid-like symptoms, not the clinical features described here. Zinc deficiency explains diarrhea, weight loss, aphthous ulcers, and perianal ulcerations, as zinc plays a

Iron would primarily manifest as anemia with fatigue and pale conjunctiva, but it does not explain the perianal and oral ulcers or chronic diarrhea. Niacin deficiency (pellagra) presents as diarrhea, dermatitis, and dementia ("3 Ds"), but it does not cause ulcerations. Selenium deficiency leads to cardiomyopathy or hypothyroid-like symptoms, not the clinical features described here. Zinc deficiency explains diarrhea, weight loss, aphthous ulcers, and perianal ulcerations, as zinc plays a role in immune function, epithelial repair, and cellular growth.

- UpToDate: "Zinc deficiency: Clinical manifestations and management."

05

A 35-year-old woman is diagnosed with iron deficiency anemia based on laboratory findings showing low hemoglobin, low MCV, and low ferritin. She is prescribed oral ferrous sulfate supplementation. What is the recommended duration of iron supplementation therapy?

- ☐ For a total of 6 months.
- ☐ For 3 months.
- ☒ For 3 months after anemia is corrected (hemoglobin returns to normal).
- ☐ Until serum iron levels normalize.

Explanation :

Iron supplementation is essential for replenishing hemoglobin and correcting the iron deficit in the bone marrow and tissues. Once hemoglobin normalizes, iron stores (reflected by ferritin) remain depleted, necessitating continued therapy to prevent relapse. Phases of Treatment:

- Correction phase: Hemoglobin normalizes within 6–8 weeks of therapy.
- Repletion phase: Therapy should continue for 3 months after normalization to replenish iron stores fully.
- Maintenance phase: Only considered in individuals at high risk for recurrent anemia (e.g., ongoing bleeding or malabsorption).

06

A 50-year-old man presents with a mass protruding from his anus during defecation. He reports intermittent rectal bleeding and discomfort. On physical examination, a soft, compressible mass is visible protruding from the anus. It reduces spontaneously after defecation. Which of the following is the most likely diagnosis?

- ☐ Anal fistula
- ☐ Anal abscess
- ☐ Anal fissure
- ☒ Hemorrhoids

Explanation :

Protruding mass during defecation suggests prolapsing internal hemorrhoids. Intermittent bright red rectal bleeding is a hallmark symptom of hemorrhoids. Internal hemorrhoids are soft, non-tender, and reduce spontaneously in early stages. Absence of systemic symptoms (no fever or severe pain) excludes more serious conditions (e.g., abscess or fistula).

Anal fistula typically presents with chronic purulent drainage or recurrent infections, not a soft, compressible mass. Anal abscess presents with acute pain, redness, swelling, and fever; a compressible mass is unlikely. Anal fissure causes sharp anal pain during defecation and bright red blood on toilet paper but no protruding mass. Hemorrhoid's classical presentation includes a protruding mass, painless bleeding, and discomfort. Internal hemorrhoids develop from dilation of the venous plexus in the rectal submucosa. In early stages, they reduce spontaneously after defecation but may progress to irreducible or thrombosed.



of the venous plexus in the rectal submucosa. In early stages, they reduce spontaneously after defecation but may progress to irreducible or thrombosed
- UpToDate: "Overview of hemorrhoids."

07

A 65-year-old male with a 30-year history of smoking presents to the emergency department with complaints of worsening shortness of breath, a persistent cough, and hemoptysis. He reports that the cough has been productive with bloody sputum for the past week. On physical examination, he is noted to have decreased breath sounds on the right side and dullness to percussion. A chest X-ray is performed and shows multiple opacities in the right lung field.

☒ Chest CT

☐ Bronchial lavage

☐ MRI of the chest

☐ Sputum culture

Explanation :

This patient presents with symptoms concerning for lung cancer, particularly with his smoking history, hemoptysis, and abnormal chest X-ray findings. A chest CT is the most appropriate next step to further evaluate the extent and nature of the lung pathology. CT imaging provides detailed visualization of the lung parenchyma, mediastinum, and surrounding structures.

- UpToDate: "Evaluation and diagnosis of hemoptysis in adults."

08

A health team in a woreda (district) aims to assess polio vaccination coverage among children under 5 years of age. They plan to conduct a house-to-house survey to collect data on vaccination status. Which of the following study designs is most appropriate for this purpose?

☐ Case series

☐ Case report

☒ Cross-sectional study

☐ Case-control study

Explanation :

The health team is assessing vaccination coverage at a single point in time. The focus is on determining the proportion of vaccinated children under 5 years of age in the district. This makes a cross-sectional study the most appropriate choice because it collects data on both exposure (vaccination) and outcome (coverage) simultaneously. Cross-sectional study is best for measuring prevalence at a given point in time. Data is collected from the target population (children under 5) through surveys. This design is efficient, cost-effective, and appropriate for public health assessments.

- Case series: Describes multiple cases with a common characteristic but does not assess prevalence or coverage.
- Case report: Describes a single case, which is irrelevant for population-based studies like vaccination coverage.
- Case-control study: Useful for identifying risk factors by comparing cases (diseased) to controls (non-diseased). This design is not appropriate for determining vaccination coverage.
- Epidemiology and Public Health Medicine, 6th Edition, UpToDate: "Study designs in epidemiology: Descriptive studies."



A 25-year-old man presents with high grade fever, left upper quadrant abdominal pain, and mild hepatosplenomegaly with tenderness on palpation. He reports recent travel to a rural area. Which of the following is the most appropriate initial diagnostic investigation?

☒ Blood culture

☐ Stool culture

☐ Bone marrow culture

☐ Stool microscopy

Explanation :

High-grade fever and left upper quadrant abdominal pain suggest systemic infection. Hepatosplenomegaly and recent travel to a rural area raise suspicion for typhoid fever or another systemic infectious disease. Blood culture is the most sensitive and specific test for identifying bacteremia and confirming the diagnosis of typhoid fever or other systemic bacterial infections. Blood culture is non-invasive and specific for systemic bacterial infections.

- UpToDate: "Typhoid fever: Clinical features, diagnosis, and treatment."

A 30-year-old man is brought to the emergency department after a motor vehicle accident. He is hypotensive (BP 90/60 mmHg), tachycardic (PR 120 bpm), and his oxygen saturation is 80% on room air. Physical examination reveals absent breath sounds on the right side of his chest. Chest X-ray shows hyperlucency and absent lung markings on the right with mediastinal shift to the left. Which of the following is the most likely diagnosis?

☐ Simple pneumothorax

☒ Tension pneumothorax

☐ Hemothorax

☐ Pulmonary contusion

Explanation :

Hypotension and tachycardia indicate compromised hemodynamics due to impaired venous return. Absent breath sounds on the right side suggests pneumothorax. Mediastinal shift is a hallmark of tension pneumothorax caused by increased intrathoracic pressure displacing mediastinal structures. Tension pneumothorax is a life-threatening emergency that requires immediate decompression via needle thoracostomy, followed by chest tube insertion. Delay in treatment can lead to cardiac arrest due to reduced venous return and hypoxemia.

• Simple pneumothorax: Characterized by absent breath sounds and hyperlucency on imaging but does not cause hemodynamic instability or mediastinal shift.

• Hemothorax: Causes dullness to percussion and absent breath sounds but does not produce hyperlucency or mediastinal shift.

• Pulmonary contusion: Typically presents with hypoxemia and patchy opacities on imaging, not hyperlucency or absent breath sounds.

- ATLS 11th, UpToDate: "Tension pneumothorax: Clinical presentation and management."

A 55-year-old woman presents to the emergency department following a fall from standing height while walking her dog. She reports experiencing severe headache that worsened progressively after the fall and has vomited twice. She is confused and disoriented, unable to recall details of the accident. On examination, she has signs of meningeal irritation (nuchal rigidity, Kernig's sign,

13

A 25-year-old man arrives at the emergency department after being shot in the lower right abdomen during a robbery. He is hypotensive and has severe abdominal pain with distension and guarding on palpation. Exploratory laparotomy (surgical exploration of the abdomen) is planned. Considering the high-velocity nature of a gunshot wound, which of the following abdominal organs is most likely to be injured?

- ☐ Liver
- ☐ Large intestine (colon)
- ☒ Small intestine
- ☐ Spleen

Explanation :

A gunshot wound to the lower right abdomen suggests high-velocity trauma. Hypotension, severe abdominal pain, and guarding are classic signs of peritoneal injury and internal bleeding. Gunshot wounds to the abdomen most commonly affect the hollow viscera (e.g., small intestine). The small intestine is the most commonly injured organ in penetrating abdominal trauma due to its length and central location in the abdomen. Management is immediate exploratory laparotomy is required to identify and repair injuries, control hemorrhage, and prevent complications like sepsis and peritonitis.

- UpToDate: "Abdominal gunshot and stab wounds: Initial evaluation and management."

14

A public health researcher wants to investigate the association between excess body weight and the development of chronic diseases like type 2 diabetes and heart disease in a large population study. Which of the following anthropometric measurements is the most widely used and considered a simple, reliable indicator of overall obesity-related health risks?

- ☒ Body mass index
- ☐ Hip circumference
- ☐ Skinfold thickness
- ☐ Waist circumference

Explanation :

The study aims to link excess body weight with chronic diseases such as type 2 diabetes and heart disease. A measurement that is simple, reliable, and widely used in large population studies is needed. BMI is the most widely used anthropometric measure for assessing overall obesity and health risks associated with excess body weight. It is calculated using weight (kg) divided by height (m^2), offering a standardized and reproducible method. Extensive epidemiological research links BMI to obesity-related health risks, including cardiovascular disease, type 2 diabetes, and mortality. BMI categories (e.g., underweight, normal weight, overweight, obese) provide a practical framework for clinical and public health applications.

15

A 52-year-old woman with a history of rheumatoid arthritis has been taking methotrexate weekly for the past two years. She presents with fatigue and reports experiencing easy bruising and frequent nosebleeds recently. Laboratory tests reveal: WBC 2,000/mm³ (low), hemoglobin 7 g/dL (anemia), MCV 110 fL (macrocytosis), and platelets 80,000/mm³ (within normal range). What is the most likely explanation for these abnormal laboratory findings?

- ☐ Active rheumatoid arthritis causing blood abnormalities

A 52-year-old woman with a history of rheumatoid arthritis has been taking methotrexate weekly for the past two years. She presents with fatigue and reports experiencing easy bruising and frequent nosebleeds recently. Laboratory tests reveal: WBC 2,000/mm³ (low), hemoglobin 7 g/dL (anemia), MCV 110 fL (macrocytosis), and platelets 80,000/mm³ (within normal range). What is the most likely explanation for these abnormal laboratory findings?

- ☐ Active rheumatoid arthritis causing blood abnormalities
- ☒ Methotrexate-induced bone marrow suppression leading to decreased blood cell production
- ☐ Vitamin B12 deficiency secondary to methotrexate use
- ☐ Myelodysplastic syndrome (MDS) developing independently of methotrexate use

Explanation :

Rheumatoid arthritis treated with methotrexate (a common disease-modifying antirheumatic drug). Methotrexate inhibits dihydrofolate reductase, interfering with folate metabolism and DNA synthesis. This can lead to bone marrow suppression, resulting in pancytopenia (affecting all three blood cell lines: WBCs, RBCs, platelets). The macrocytosis (MCV 110 fL) is characteristic of folate deficiency due to methotrexate. Management is based on immediate folate supplementation (e.g., folinic acid or folic acid) and discontinuation of methotrexate may be necessary, depending on the severity of bone marrow suppression.

- UpToDate: "Hematologic toxicity of methotrexate in rheumatoid arthritis."

A 35-year-old male presents to the clinic with a 3-month history of cough, shortness of breath, and night sweats. He has a history of pulmonary tuberculosis (TB) treated 4 years ago. Upon examination, he appears cachectic and has decreased breath sounds on auscultation. A chest X-ray shows bilateral infiltrates and possible cavitory lesions. The patient is also found to be HIV positive with a CD4 count of 350 cells/mm³.

- ☐ Full course of TB treatment
- ☐ Isoniazid preventive therapy (IPT) after ruling out active TB
- ☐ Initiate antiretroviral therapy (ART)
- ☒ Obtain a sputum culture for acid-fast bacilli

Explanation :

The patient has clinical features of recurrent tuberculosis (TB) with HIV coinfection, a history of prior TB treatment, and concerning symptoms such as cough, night sweats, and cavitory lesions. The first step is to confirm active TB through a sputum culture for acid-fast bacilli (AFB) or other diagnostic tests such as nucleic acid amplification testing (NAAT). Diagnosis must precede treatment to guide appropriate therapy and drug resistance testing.

Initiating TB treatment without microbiological confirmation risks over-treatment or inappropriate treatment, especially in a patient with prior TB and potential drug resistance. Isoniazid preventive therapy (IPT) is for latent TB and not appropriate in a patient with signs of active TB. ART is critical in HIV patients, it should not be initiated until TB has been adequately assessed and treated to avoid immune reconstitution inflammatory syndrome (IRIS).

- UpToDate

A medical student is performing a neurological examination. To elicit the plantar reflex, the lateral aspect of the sole of the foot is stroked. Which of the following nerve root levels is primarily

17

A medical student is performing a neurological examination. To elicit the plantar reflex, the lateral aspect of the sole of the foot is stroked. Which of the following nerve root levels is primarily responsible for mediating the normal plantar reflex (plantar flexion of the toes)?

☒ S1-S2

☐ S2-S3

☐ S3-S4

☐ L4-L5

Explanation :

The plantar reflex is elicited by stroking the lateral aspect of the sole of the foot.

A normal response involves plantar flexion of the toes (downward movement). The plantar reflex is mediated by the tibial nerve, a branch of the sciatic nerve, which primarily originates from S1-S2 nerve roots. These nerve roots contribute to the motor function of muscles responsible for plantar flexion.

S2-S4 are primarily associated with pelvic and perineal innervation, not the plantar reflex. L4-L5 are involved in knee extension (L4) and dorsiflexion of the foot (L5), not plantar flexion. The Babinski sign, an abnormal plantar reflex where the toes extend upward, indicates dysfunction of the corticospinal tract (upper motor neuron lesion). Testing the plantar reflex helps assess both peripheral nerve integrity (S1-S2) and central nervous system function.

- UpToDate: "Neurologic examination in adults."

18

A 25-year-old man presents with acute onset of quadriplegia and absent deep tendon reflexes. He also reports recent onset of urinary retention. His vital signs are stable. Laboratory investigations reveal a white blood cell count (WBC) of 8,000/mm³. Cerebrospinal fluid (CSF) analysis shows 30 cells/mm³ (lymphocytic pleocytosis) and a protein level of 6 g/L (markedly elevated). Which of the following is the most likely initial treatment for this patient?

☐ Intravenous ceftriaxone

☒ Intravenous immunoglobulin (IVIG)

☐ Intravenous methylprednisolone

☐ Intravenous acyclovir

Explanation :

Acute quadriplegia with absent deep tendon reflexes: Suggestive of a lower motor neuron lesion (peripheral neuropathy). Urinary retention indicates autonomic nervous system involvement. CSF findings of lymphocytic pleocytosis (30 cells/mm³) with markedly elevated protein (6 g/L). This is consistent with albuminocytologic dissociation, a hallmark of GBS. GBS is an acute autoimmune neuropathy, often triggered by infections (e.g., *Campylobacter jejuni*).

Treatment options are IVIG or plasmapheresis are first-line therapies. Both reduce autoantibody-mediated nerve damage and improve recovery time. IVIG is often preferred due to ease of administration and fewer complications compared to plasmapheresis. Early treatment reduces the risk of respiratory failure and long-term disability.

- UpToDate: "Guillain-Barré syndrome in adults: Pathogenesis, clinical features, and diagnosis."

19

A surgeon is preparing to perform an inguinal herniorrhaphy. Which of the following steps is considered the most crucial for preventing surgical site infection?



A surgeon is preparing to perform an inguinal herniorrhaphy. Which of the following steps is considered the most crucial for preventing surgical site infection?

- ☒ Thorough handwashing with antiseptic soap
- ☐ Application of povidone-iodine to the surgical site
- ☐ Preoperative shaving of the surgical site
- ☐ Wearing sterile surgical gloves

Explanation :

Surgical site infections (SSIs) are among the most common post-operative complications. Prevention requires strict adherence to aseptic techniques. Handwashing with antiseptic soap eliminates transient and reduces resident microorganisms on the surgeon's hands. Contaminated hands are a major source of infection, and handwashing is the cornerstone of infection control in the operating room. Even with gloves, micro-perforations can occur during surgery, making hand hygiene essential.

- Surgical Infections: Principles and Practice, 2nd Edition, UpToDate: "Strategies to prevent surgical site infections in adults."

A 40-year-old woman presents with a swelling in her right groin area that she noticed several months ago. She reports that the swelling becomes more prominent when she coughs or strains. On physical examination, a soft, reducible mass is palpated below the inguinal ligament. Which of the following types of hernia is most likely?

- ☐ Direct inguinal hernia
- ☒ Femoral hernia
- ☐ Indirect inguinal hernia
- ☐ Obturator hernia

Explanation :

Right groin swelling: Chronic and reducible, typical of hernias. Swelling below the inguinal ligament: This anatomical location strongly suggests a femoral hernia. Becomes more prominent with coughing or straining: Indicates increased intra-abdominal pressure pushing the hernia outward. Femoral hernias protrude through the femoral ring, located below the inguinal ligament and medial to the femoral vein. They are more common in women due to a wider pelvis and can often present with complications like incarceration or strangulation.

- Direct inguinal hernia: Protrudes through the posterior wall of the inguinal canal and lies above the inguinal ligament.
 - Indirect inguinal hernia: Passes through the inguinal canal and into the scrotum or labia, also located above the inguinal ligament.
 - Obturator hernia: Rare and occurs through the obturator canal. Often presents with Howship-Romberg sign (pain radiating to the medial thigh).
- Bailey 27th, UpToDate: "Clinical features, diagnosis, and management of groin hernias in adults."

A 25-year-old man was admitted to the orthopedic ward after sustaining a closed fracture of his tibia and fibula in a motorcycle accident. Three days post-operatively, he reports severe, unrelenting pain in his lower leg, disproportionate to his injury. On examination, his lower leg is markedly



21

A 25-year-old man was admitted to the orthopedic ward after sustaining a closed fracture of his tibia and fibula in a motorcycle accident. Three days post-operatively, he reports severe, unrelenting pain in his lower leg, disproportionate to his injury. On examination, his lower leg is markedly swollen, tense, and exquisitely tender to palpation. Distal pulses (dorsalis pedis and posterior tibial) are not palpable. There is also paresthesia in the foot. Which of the following is the most appropriate next step in management?

☐ Measuring compartment pressures

☒ Urgent fasciotomy

☐ Fracture fixation revision

☐ Administration of intravenous analgesics

Explanation :

Compartment syndrome is a surgical emergency. Compartment syndrome occurs when increased pressure within a closed fascial compartment compromises blood flow and nerve function. Most common after fractures or trauma, especially in the tibia or forearm. Severe pain disproportionate to the injury, swollen, tense lower leg with tenderness, absent distal pulses and paresthesia are indicators of compartment syndrome. Urgent fasciotomy is required to release the pressure and restore perfusion, preventing ischemia and necrosis of muscles and nerves. Delayed treatment can lead to Volkmann's contracture or limb loss.

- Campbell's Operative Orthopaedics, 14th, UpToDate: "Acute compartment syndrome of the extremities."

22

An 18-year-old female presents to the clinic with complaints of dysuria and increased urinary frequency for the past two weeks. She denies any fever, flank pain, or vaginal discharge. A urinalysis is performed, which shows no evidence of infection (negative nitrites and leukocyte esterase). The urine test is otherwise normal, with no signs of hematuria or proteinuria.

☐ Urine culture

☐ Fosfomycin

☐ Reassure and send home

☒ Ultrasound of the kidneys and bladder

Explanation :

This 18-year-old female presents with persistent dysuria and increased urinary frequency without evidence of infection on urinalysis (negative nitrites and leukocyte esterase). Non-invasive imaging to evaluate structural abnormalities, such as bladder outlet obstruction, hydronephrosis, or urinary tract stones.

It is appropriate when symptoms persist despite normal urinalysis.

- UpToDate

23

In some traditional communities, a practice is performed on newborns shortly after birth with the intention of preventing feeding problems. This practice involves the removal of the uvula. What is the name of this traditional practice?

☐ Milk tooth extraction

☒ Uvulectomy

☐ Tongue-tie clipping



In some traditional communities, a practice is performed on newborns shortly after birth with the intention of preventing feeding problems. This practice involves the removal of the uvula. What is the name of this traditional practice?

☐ Milk tooth extraction

☒ Uvulectomy

☐ Tongue-tie clipping

☐ Palatoplasty

Explanation :

Traditional practice in some cultures, often performed on neonates or young children. Believed to prevent feeding or respiratory problems. The practice lacks scientific support and carries risks of complications such as bleeding, infection, and airway obstruction. Uvulectomy refers to the surgical removal of the uvula, the small fleshy tissue hanging from the soft palate at the back of the throat. Practiced in some regions of Africa and the Middle East as part of traditional medicine.

- Nelson 21st, UpToDate: "Cultural practices and beliefs in pediatric care."

A 30-year-old woman is brought to the emergency department after a motor vehicle collision. She has significant chest trauma and is experiencing respiratory distress. On examination, she is coughing up blood-tinged secretions and has a large open wound on the right side of her chest that makes a sucking sound with each breath. Which of the following is the most appropriate *initial* management step?

☒ Application of a three-sided occlusive dressing to the chest wound

☐ Suctioning of airway secretions

☐ Administration of supplemental oxygen

☐ Insertion of a chest tube

Explanation :

Open chest wound producing a sucking sound with each breath. Creates a one-way valve effect, allowing air to exit during expiration while preventing air entry during inspiration. It prevents worsening of the pneumothorax and allows stabilization until definitive care (e.g., chest tube insertion) is provided.

- ATLS 10th, UpToDate: "Thoracic trauma: Initial evaluation and management."

A 45-year-old woman underwent a total thyroidectomy for multinodular goiter. On the first postoperative day, she complains of numbness and tingling around her mouth and in her hands. Physical examination reveals a positive Chvostek's sign and Trousseau's sign. Which of the following is the most appropriate treatment?

☐ Oral calcium supplementation

☒ Intravenous calcium gluconate

☐ Oral vitamin D supplementation

☐ Intramuscular calcitonin

Explanation :

Perioral numbness, tingling in hands, positive Chvostek's sign (facial twitching when tapping



A 45-year-old woman underwent a total thyroidectomy for multinodular goiter. On the first postoperative day, she complains of numbness and tingling around her mouth and in her hands. Physical examination reveals a positive Chvostek's sign and Trousseau's sign. Which of the following is the most appropriate treatment?

- ☐ Oral calcium supplementation
- ☒ Intravenous calcium gluconate
- ☐ Oral vitamin D supplementation
- ☐ Intramuscular calcitonin

Explanation :

Perioral numbness, tingling in hands, positive Chvostek's sign (facial twitching when tapping the facial nerve) and Trousseau's sign (carpopedal spasm induced by blood pressure cuff) are symptoms of hypocalcemia. History of total thyroidectomy suggests accidental injury or removal of the parathyroid glands, leading to transient or permanent hypoparathyroidism. Hypoparathyroidism causes low parathyroid hormone (PTH) levels, leading to:

- Reduced calcium resorption from bones.
- Decreased calcium reabsorption and increased phosphate reabsorption in the kidneys.
- Impaired activation of vitamin D, reducing calcium absorption in the gut.

Severe or symptomatic hypocalcemia (e.g., neuromuscular irritability) requires immediate correction with IV calcium gluconate to avoid complications like seizures, laryngospasm, or cardiac arrhythmias. Once stabilized, oral calcium and vitamin D supplementation can be initiated for long-term management.

- UpToDate: "Postoperative hypocalcemia following thyroid surgery."

A researcher is studying the age distribution of patients diagnosed with a specific condition. The ages of 12 patients are recorded as follows: 12, 19, 23, 55, 55, 54, 56, 51, 52, 53, 54, 56. Which measure of central tendency best represents the typical age of patients in this sample, minimizing the influence of outliers?

- ☐ Arithmetic mean
- ☐ Mode
- ☒ Median
- ☐ Range

Explanation :

The median is less influenced by outliers compared to the mean and provides a more accurate representation of central tendency for skewed data. The median is the middle value in an ordered dataset:

- Ordered data: 12, 19, 23, 51, 52, 53, 54, 54, 55, 55, 56, 56.
 - Median = (6th value + 7th value) ÷ 2 = (53 + 54) ÷ 2 = 53.5.
- It minimizes the effect of extreme values (e.g., 12 and 19).

- Principles of Biostatistics, 2nd

A 25-year-old male presents to the ED after a fall from his bike. He complains of severe pain in his forearm. An X-ray reveals fractures in the lower third and upper third of the forearm. On physical examination, there is swelling and tenderness over the fracture sites. The patient has difficulty extending his wrist and fingers, and there is sensory loss over the dorsal aspect of the hand.



A 25-year-old male presents to the ED after a fall from his bike. He complains of severe pain in his forearm. An X-ray reveals fractures in the lower third and upper third of the forearm. On physical examination, there is swelling and tenderness over the fracture sites. The patient has difficulty extending his wrist and fingers, and there is sensory loss over the dorsal aspect of the hand.

☐ Lateral cutaneous nerve

☒ Radial nerve

☐ Ulnar nerve

☐ Median nerve

Explanation :

The radial nerve is the most likely to be injured with fractures of the lower and upper thirds of the forearm. This nerve runs along the posterior aspect of the humerus and then crosses over to the lateral side of the forearm, providing motor function to the extensor muscles of the wrist and fingers and sensory function to the dorsal aspect of the hand. The described symptoms of difficulty in wrist and finger extension (wrist drop) and sensory loss over the dorsal aspect of the hand are characteristic of radial nerve injury.

Lateral cutaneous nerve (of forearm) is a continuation of musculocutaneous nerve and its injury affects sensation over the lateral forearm.

The ulnar nerve typically affects the medial aspect of the hand and provides motor innervation to the intrinsic muscles of the hand, leading to symptoms such as claw hand and sensory loss over the fifth finger and medial half of the fourth finger. It is less commonly injured in forearm fractures compared to the radial nerve.

The median nerve provides motor innervation to the flexor muscles of the forearm and some intrinsic muscles of the hand. Injury to this nerve would result in symptoms such as difficulty with thumb opposition and sensory loss over the palmar aspect of the first three and a half fingers. The symptoms described in the case are not consistent with median nerve injury.

- Apley 10th p295, Tintinalli's 9th p1833, p1813, p1818

A mother brings her 4-month-old baby to the clinic with complaints of photophobia and tearing after the baby accidentally poked her right eye with his finger while playing. On examination, the affected eye appears red and she reports discomfort exacerbated by blinking. Visual acuity is normal, and there is no discharge or foreign body visible on the conjunctiva. What is the most likely diagnosis?

☒ Corneal abrasion

☐ Conjunctival injury

☐ Foreign body

☐ Iritis

Explanation :

A corneal abrasion is a common injury resulting from mechanical trauma to the corneal surface, such as a finger poke. Symptoms include photophobia (sensitivity to light), tearing, foreign body sensation, and discomfort exacerbated by blinking. The redness of the eye and the acute onset of symptoms are consistent with a corneal abrasion. Work-up includes Slit-lamp examination, which uses fluorescein dye, measure the size (e.g., height and width) of the abrasion, and diagram its location. Evaluate for an AC reaction, infiltrate (underlying corneal opacification), corneal laceration, and penetrating trauma.

Conjunctival injury can cause redness and discomfort, it typically presents with less severe symptoms of photophobia and tearing compared to a corneal abrasion. In this scenario, the symptoms are more indicative of corneal involvement due to the direct trauma to the cornea.

A foreign body in the eye can cause similar symptoms, but the history of a poke by the baby's finger makes corneal abrasion more likely than a retained foreign body. Foreign bodies often present with a sensation of irritation or grittiness but may not cause as much photophobia unless they are lodged in the cornea.

Iritis is inflammation of the iris and can cause photophobia and pain, but it is less likely to be caused directly by trauma such as a finger poke. Additionally, iritis typically presents with other signs such as a small, irregular pupil and pain that worsens with eye movement, which are not described in this case.

- The Wills Eye Manual 6th p16, UpToDate

29

A hospital manager conducts a training program for nurses on the proper use of the partograph during labor. After the training, a post-training assessment is administered to evaluate the nurses' knowledge and skills. The results indicate that the training was effective in improving their understanding and application of the partograph. Which aspect of program evaluation is being assessed?

☐ Process evaluation

☐ Impact evaluation

☒ Outcome evaluation

☐ Needs assessment

Explanation :

The training program aimed to improve nurses' knowledge and application of the partograph. Post-training assessment indicates an improvement in their understanding and practical application. Outcome evaluation measures the effectiveness of a program in achieving its intended results (e.g., improved skills or knowledge). It assesses what changed as a result of the intervention (e.g., increased competency in using the partograph).

- **Process evaluation:** Focuses on the implementation of the program (e.g., whether the training was delivered as planned, participant attendance). It does not measure the effectiveness of the intervention.
- **Impact evaluation:** Assesses the long-term effects of the program on broader outcomes (e.g., reduced maternal and neonatal mortality due to better partograph use). This scenario does not assess long-term outcomes.

30

A 28-year-old man presents to the emergency department after a motor vehicle accident. He sustained blunt trauma to his abdomen and reports blood in his urine. He is hemodynamically stable. Which of the following is the most appropriate initial diagnostic investigation to evaluate for genitourinary injury?

☐ Abdominal ultrasound

☐ Cystourethrography (retrograde or voiding)

☒ CT scan of the abdomen and pelvis with intravenous contrast

Intravenous urogram (IIV)



A 28-year-old man presents to the emergency department after a motor vehicle accident. He sustained blunt trauma to his abdomen and reports blood in his urine. He is hemodynamically stable. Which of the following is the most appropriate initial diagnostic investigation to evaluate for genitourinary injury?

- ☐ Abdominal ultrasound
- ☐ Cystourethrography (retrograde or voiding)
- ☒ CT scan of the abdomen and pelvis with intravenous contrast
- ☐ Intravenous pyelogram (IVP)

Explanation :

Blunt abdominal trauma with hematuria suggests potential genitourinary (GU) injury (e.g., kidney, ureters, bladder). The patient is hemodynamically stable, making advanced imaging feasible. CT scan with IV contrast is the gold standard for evaluating GU injuries in hemodynamically stable patients. Provides detailed visualization of renal parenchyma, collecting system, ureters, bladder, and surrounding structures. Allows detection of renal lacerations, contusions, perinephric hematomas, and vascular injuries. Enables accurate grading of renal injuries (e.g., per AAST guidelines) and aids in deciding between conservative vs. surgical management.
- Smith's General Urology, 18th Edition.

31

Researchers are conducting a study on the prevalence of depression in a population with high literacy rates. They need to collect detailed information about participants' symptoms, mood, and experiences related to depression. Which of the following data collection methods is most appropriate for this population?

- ☐ Face-to-face interviews
- ☒ Self-administered questionnaires
- ☐ Focus group discussions
- ☐ Direct observation

Explanation :

The target population has high literacy rates, meaning they can effectively understand and respond to written materials. Researchers need detailed, individualized information about depression symptoms.
- Research Methods in Psychology, 4th

32

Two physicians in a busy emergency department have differing opinions on the optimal management of a critically ill patient. One physician advocates for aggressive intervention, while the other prefers a more conservative approach. They engage in a discussion, actively listening to each other's perspectives and ultimately reaching a mutually agreeable treatment plan that incorporates elements of both approaches. Which conflict management style did they employ?

- ☐ Competing
- ☐ Accommodating
- ☐ Compromising
- ☒ Collaborating

Explanation :



Two physicians in a busy emergency department have differing opinions on the optimal management of a critically ill patient. One physician advocates for aggressive intervention, while the other prefers a more conservative approach. They engage in a discussion, actively listening to each other's perspectives and ultimately reaching a mutually agreeable treatment plan that incorporates elements of both approaches. Which conflict management style did they employ?

- ☐ Competing
- ☐ Accommodating
- ☐ Compromising
- ☒ Collaborating

Explanation :

The two physicians actively listened to each other's viewpoints and incorporated elements of both perspectives to develop an optimal treatment plan. Collaboration emphasizes open communication and mutual respect to achieve a solution that satisfies all parties.

- **Competing:** One physician imposes their solution without regard for the other's perspective. This is adversarial and does not align with the described scenario.
- **Accommodating:** One physician prioritizes the other's preferences over their own, which does not describe the mutual agreement achieved here.
- **Compromising:** While compromise seeks a middle ground, it often involves sacrificing some interests. In contrast, collaboration seeks a win-win solution.

A public health worker aims to provide education about safe abortion practices in a rural community with limited access to healthcare facilities and varying levels of literacy. Which of the following communication strategies is likely to be most effective in this setting?

- ☐ Distributing pamphlets and brochures
- ☐ Broadcasting radio announcements
- ☒ Conducting interpersonal communication
- ☐ Utilizing online educational platforms

Explanation :

The target population has limited healthcare access and varying literacy levels. Effective communication must overcome language, literacy, and cultural barriers while fostering trust. Interpersonal communication strategies (e.g., small group discussions or community meetings) enable direct engagement with the audience. Facilitates two-way communication, allowing the public health worker to clarify doubts, correct misinformation, and adapt messaging based on feedback. Builds trust and rapport, particularly important in sensitive topics like abortion. Provides the flexibility to tailor information to the cultural context and literacy level of the community.

A hospital administrator makes decisions independently, rarely consulting with staff or seeking their input. They enforce strict rules and procedures, expecting strict adherence from all employees. Which leadership style is this administrator exhibiting?

- ☐ Transformational leadership
- ☐ Democratic leadership



A hospital administrator makes decisions independently, rarely consulting with staff or seeking their input. They enforce strict rules and procedures, expecting strict adherence from all employees. Which leadership style is this administrator exhibiting?

- ☐ Transformational leadership
- ☐ Democratic leadership
- ☐ Laissez-faire leadership
- ☒ Autocratic leadership

Explanation :

The hospital administrator rarely consults staff or seeks input. Strict adherence to rules is expected, with little room for flexibility or collaboration. Autocratic leadership involves centralized decision-making, where the leader makes decisions independently and enforces strict rules without consulting team members. Prioritizes control and efficiency but often undermines staff morale and creativity. Suitable for emergencies requiring rapid decisions but may lead to dissatisfaction in long-term, non-crisis settings.

- Transformational leadership: Focuses on inspiring and motivating employees to achieve shared goals, which is not evident here.
- Democratic leadership: Emphasizes collaboration and input from team members, contrasting with the administrator's independent decision-making.
- Laissez-faire leadership: Characterized by minimal supervision and delegation, which is the opposite of the strict enforcement described.
- Leadership in Healthcare: Essential Values and Skills, UpToDate: "Styles of leadership and their impact on healthcare settings."

In a team meeting, a manager consistently criticizes and blames team members for any errors or setbacks. They create a hostile and tense atmosphere, discouraging open communication and collaboration. Which of the following negative group roles is this manager demonstrating?

- ☒ The aggressor
- ☐ The blocker
- ☐ The recognition-seeker
- ☐ The self-confessor

Explanation :

The manager criticizes and blames team members. Their behavior fosters a hostile and tense work environment, stifling collaboration. The aggressor role involves criticizing or blaming others, often without justification, undermining team morale and creating a toxic atmosphere, discouraging open communication by instilling fear of judgment. Aggressors hinder team dynamics and can cause long-term damage to employee performance and well-being.

- Blocker: Resists progress by being overly critical or stubborn but doesn't necessarily create a hostile environment.
- Recognition-seeker: Focuses on drawing attention to their achievements, which is unrelated to the described behavior.
- Self-confessor: Shares personal issues or unrelated opinions during team activities, which doesn't align with creating hostility.
- UpToDate: "Workplace conflict and its impact on team performance."

A 62-year-old woman presents to the emergency department with progressive jaundice, dark urine, and light-colored stools for the past 3 weeks. Her skin exhibits a yellowish discoloration, and multiple ecchymoses are noted on her arms and legs. She reports no abdominal pain but has experienced significant fatigue and weight loss. Her past medical history is significant for hypertension and hyperlipidemia, managed with medications. On physical examination, her sclerae are icteric, and her abdomen is soft with mild right upper quadrant tenderness. Laboratory studies reveal elevated total and direct bilirubin, alkaline phosphatase, and gamma-glutamyl transferase (GGT). Her prothrombin time (PT) and international normalized ratio (INR) are prolonged. Which blood component is MOST appropriate to administer before a planned biliary drainage procedure?

- ☐ Whole blood
- ☒ Fresh frozen plasma
- ☐ Platelets
- ☐ Packed red blood cells

Explanation :

The patient has obstructive jaundice with prolonged PT/INR, indicating a coagulation defect due to vitamin K deficiency from impaired bile flow. Biliary drainage is an invasive procedure, which carries a risk of bleeding in a coagulopathic patient. FFP contains clotting factors that can rapidly correct coagulopathy caused by a deficiency of vitamin K-dependent factors (factors II, VII, IX, X). It is the most effective and immediate option for addressing coagulation abnormalities in preparation for procedures. Administering FFP reduces bleeding risks during biliary drainage procedures.

- UpToDate: "Management of coagulopathy in liver disease."

A 43-year-old male presents to your office with several months history of progressive fatigue and occasional palpitations. He had three episodes of acute respiratory infections over the last year; otherwise, his past medical history is insignificant. He smokes two packs of cigarettes daily and consumes alcohol occasionally. He is not taking any medications and denies drug abuse. He has been sexually active with several partners recently. The physical examination is insignificant. CBC shows Hb 6.7 g/dL, RBC 2.3 million/cmm, WBC 5,900/cmm, Segmented neutrophils 60%, Bands 3%, Lymphocytes 30% and MCV 85 fL. His HIV test results are negative. Which of the following is the best next step in the management of this patient?

- ☐ ESR and serum C-reactive protein
- ☐ Serum iron and TIBC
- ☒ Reticulocyte count
- ☐ Hemoglobin electrophoresis and Coombs' test

Explanation :

A 3-day-old neonate is brought to the emergency department due to difficulty feeding and episodes of choking and coughing during breastfeeding. The mother reports excessive drooling. On examination, the infant is tachypneic and has mild intercostal retractions. Auscultation of the chest reveals a holosystolic murmur best heard at the left lower sternal border. The infant's oxygen saturation is 92% on room air. An orogastric tube has been placed for feeding. Which of the following is the MOST appropriate next diagnostic step?



A 3-day-old neonate is brought to the emergency department due to difficulty feeding and episodes of choking and coughing during breastfeeding. The mother reports excessive drooling. On examination, the infant is tachypneic and has mild intercostal retractions. Auscultation of the chest reveals a holosystolic murmur best heard at the left lower sternal border. The infant's oxygen saturation is 92% on room air. An orogastric tube has been placed for feeding. Which of the following is the MOST appropriate next diagnostic step?

☒ Echocardiogram

☐ Chest X-ray

☐ Abdominal ultrasound

☐ Head CT scan

Explanation :

Feeding difficulty, choking, and excessive drooling raise suspicion for tracheoesophageal fistula (TEF) or other congenital anomalies. Presence of a holosystolic murmur and desaturation indicates possible cardiac involvement, as congenital heart defects often co-occur with TEF. An echocardiogram is critical to evaluate for cardiac defects, which are common in neonates with suspected TEF (e.g., part of the VACTERL association). Identifying congenital heart disease is vital before any surgical intervention for TEF to manage anesthesia and perioperative risks. Chest X-ray may show signs of TEF (e.g., gas-filled stomach or misplaced orogastric tube), but it does not evaluate the heart.

- Nelson 21st

A 9-year-old boy presents to the clinic with a 3-day history of abdominal pain. The pain initially started in the periumbilical region but has become more diffuse over the past 24 hours. He has also developed fever, nausea, and anorexia. On physical examination, his temperature is 101°F (38.3°C), and his heart rate is 120 bpm. His abdomen is tender to palpation, with guarding and rebound tenderness in the right lower quadrant. What is the MOST likely diagnosis?

☐ Gastroenteritis

☐ Mesenteric adenitis

☒ Appendicitis

☐ Intussusception

Explanation :

Appendicitis is the most common surgical emergency in children. The migration of pain and localized peritoneal irritation are hallmark findings. Diagnosis is often clinical, but imaging (e.g., ultrasound or CT) may confirm.

A 55-year-old man presents to the clinic complaining of chest pain that occurs during exertion. He describes the pain as a squeezing sensation in the center of his chest, which radiates to his left arm. The pain is relieved by rest within a few minutes. He has a history of hypertension and hyperlipidemia, for which he takes medications. He is a smoker with a 20-pack-year history. Which of the following is the BEST initial diagnostic test for this patient?

☒ Electrocardiogram (ECG)

☐ Exercise stress test

A 55-year-old man presents to the clinic complaining of chest pain that occurs during exertion. He describes the pain as a squeezing sensation in the center of his chest, which radiates to his left arm. The pain is relieved by rest within a few minutes. He has a history of hypertension and hyperlipidemia, for which he takes medications. He is a smoker with a 20-pack-year history. Which of the following is the BEST initial diagnostic test for this patient?

☒ Electrocardiogram (ECG)

☐ Exercise stress test

☐ Coronary angiography

☐ Chest X-ray

Explanation :

ECG is the first-line test for patients with chest pain to rule out ischemic changes or arrhythmias. Resting ECG may show ST depression, T wave inversion, or be normal in stable angina. Further testing (e.g., stress testing or coronary angiography) depends on ECG findings and clinical suspicion.

Exercise stress test is appropriate if the resting ECG is normal and clinical suspicion for CAD remains. Coronary angiography is a definitive diagnostic test for CAD but is invasive and typically reserved for patients with high-risk findings. Chest X-ray is useful for structural lung or mediastinal abnormalities but does not evaluate coronary arteries.

- UpToDate: "Stable ischemic heart disease: Overview of diagnosis."

A 78-year-old woman with a history of hypertension and coronary artery disease presents to the emergency department with palpitations and shortness of breath. She reports feeling her heart racing for the past several hours. On examination, her heart rate is 160 bpm and irregularly irregular. Her blood pressure is 100/60 mmHg. An ECG reveals atrial fibrillation with rapid ventricular response. Which of the following is the MOST appropriate initial treatment for this patient?

☐ Amiodarone

☒ Metoprolol

☐ Digoxin

☐ Electrical cardioversion

Explanation :

Symptoms of palpitations and shortness of breath with tachycardia (HR: 160 bpm) and irregularly irregular rhythm are consistent with atrial fibrillation (AF). BP of 100/60 mmHg suggests relative hemodynamic stability despite the rapid ventricular response (RVR). Beta-blockers (e.g., metoprolol) are first-line for rate control in hemodynamically stable patients with AF and RVR. Metoprolol reduces ventricular rate by blocking AV node conduction and is especially effective in patients with preserved LV function.

Amiodarone is used for rhythm control, not first-line for acute rate control in hemodynamically stable patients. Digoxin is effective for rate control (especially in presence of acute heart failure) but acts slowly and is less effective during exertion or high adrenergic states. Electrical cardioversion is indicated in hemodynamically unstable patients or for rhythm control if AF onset is <48 hours.

- UpToDate: "Management of atrial fibrillation: Rate control and rhythm control strategies."

42

A 14-year-old boy presents to his pediatrician with periorbital and lower extremity edema. His mother reports that his urine has been foamy for the past few weeks. He denies any recent infections or illnesses. On physical examination, he has significant pitting edema in his lower extremities and periorbital puffiness. Urinalysis reveals 4+ proteinuria and microscopic hematuria. Serum albumin is low. Which of the following is the MOST likely diagnosis?

☐ Focal segmental glomerulosclerosis

☒ Minimal change disease

☐ Membranoproliferative glomerulonephritis

☐ IgA nephropathy

Explanation :

Minimal Change Disease is most common cause of nephrotic syndrome in this age group. Diagnosis is often clinical, but a kidney biopsy (if performed) would show podocyte effacement on electron microscopy with no changes on light microscopy. Responds well to corticosteroids.
- Nelson 21st, UpToDate: "Minimal change disease in children: Clinical features and diagnosis."

43

A 60-year-old man with type 2 diabetes mellitus is being managed with metformin and a sulfonylurea. His HbA1c is 7.5%. His lipid panel reveals an LDL cholesterol of 110 mg/dL, HDL cholesterol of 35 mg/dL, and triglycerides of 200 mg/dL. He has a history of hypertension and is a current smoker. Which of the following medications would be MOST beneficial in reducing his cardiovascular risk?

☐ Ezetimibe

☐ Fibrate

☒ Statin

☐ Bile acid sequestrant

Explanation :

Statins are first-line therapy for cardiovascular risk reduction in diabetics, irrespective of LDL level, due to their pleiotropic effects (reduce LDL, stabilize plaques, and anti-inflammatory benefits). Guidelines recommend moderate- to high-intensity statins for diabetics aged 40–75 with LDL $\geq$ 70 mg/dL. Being diabetic automatically places this patient in a high cardiovascular risk group. Also LDL cholesterol of 110 mg/dL and HDL of 35 mg/dL warrant intervention to lower cardiovascular risk.

- Ezetimibe: A second-line option to lower LDL but is less effective than statins for reducing cardiovascular risk.
- Fibrate: Primarily used for hypertriglyceridemia (>500 mg/dL) to prevent pancreatitis, not first-line for LDL management.
- Bile acid sequestrant: Reduces LDL but is less effective than statins and poorly tolerated due to GI side effects.
- UpToDate: "Management of hyperglycemia in type 2 diabetes: Cardiovascular risk reduction."

44

A 30-year-old man presents to the emergency department with high-grade fever, chills, and rigors. He recently returned from a trip to sub-Saharan Africa. He is lethargic and pale. Laboratory studies reveal a creatinine of 3.2 mg/dL, hemoglobin of 6 g/dL, and a positive blood smear for *Plasmodium falciparum*. Which of the following is the MOST appropriate treatment for this patient?



44

A 30-year-old man presents to the emergency department with high-grade fever, chills, and rigors. He recently returned from a trip to sub-Saharan Africa. He is lethargic and pale. Laboratory studies reveal a creatinine of 3.2 mg/dL, hemoglobin of 6 g/dL, and a positive blood smear for *Plasmodium falciparum*. Which of the following is the MOST appropriate treatment for this patient?

☐ Oral atovaquone-proguanil

☒ Intravenous artesunate

☐ Oral chloroquine

☐ Oral doxycycline

Explanation :

High-grade fever, chills, and rigors: Malaria symptoms. Recent travel to sub-Saharan Africa: Endemic area for *Plasmodium falciparum*, often resistant to chloroquine. The patient has signs of severe malaria (Scr 3.2 mg/dL (AKI), hemoglobin 6 g/dl, and lethargy. IV artesunate is the gold standard for severe falciparum malaria, as recommended by the WHO. It rapidly clears parasitemia and prevents complications such as cerebral malaria. Oral antimalarials (e.g., atovaquone-proguanil or chloroquine) are insufficient for severe disease.

• Oral atovaquone-proguanil is effective for uncomplicated malaria but inappropriate for severe cases.

45

A 45-year-old woman presents with progressive shortness of breath on exertion and orthopnea. She reports a history of rheumatic fever as a child. On physical examination, a loud S1 and an opening snap followed by a low-pitched diastolic murmur are heard at the apex. An echocardiogram reveals an ejection fraction of 65%, a mitral valve area of 1.5 cm², and thickened mitral valve leaflets. What is the MOST likely diagnosis?

☐ Mitral regurgitation

☒ Mitral stenosis

☐ Aortic stenosis

☐ Tricuspid stenosis

Explanation :

46

A 50-year-old woman who underwent laparoscopic cholecystectomy 3 days ago presents to the emergency department with severe epigastric pain radiating to the back. She also reports nausea and vomiting. On examination, her abdomen is tender to palpation in the epigastric region. Laboratory studies reveal elevated serum amylase and lipase levels, eight times the upper limit of normal. What is the MOST appropriate initial management for this patient?

☐ Immediate surgical exploration

☐ Endoscopic retrograde cholangiopancreatography (ERCP)

☒ Conservative management with bowel rest, intravenous fluids, and analgesics

☐ Administration of antibiotics

Explanation :



A 50-year-old woman who underwent laparoscopic cholecystectomy 3 days ago presents to the emergency department with severe epigastric pain radiating to the back. She also reports nausea and vomiting. On examination, her abdomen is tender to palpation in the epigastric region. Laboratory studies reveal elevated serum amylase and lipase levels, eight times the upper limit of normal. What is the MOST appropriate initial management for this patient?

- ☐ Immediate surgical exploration
- ☐ Endoscopic retrograde cholangiopancreatography (ERCP)
- ☒ Conservative management with bowel rest, intravenous fluids, and analgesics
- ☐ Administration of antibiotics

Explanation :

Gallstone-related biliary pancreatitis is a common cause of post-surgical acute pancreatitis. Severe epigastric pain radiating to the back, nausea, and vomiting are classic. Laboratory Findings of elevated serum amylase and lipase (≥ 3 times the upper limit of normal) confirm the diagnosis. Initial Management are bowel rest, intravenous fluids and analgesics. Most cases of mild to moderate acute pancreatitis resolve with conservative treatment.

- UpToDate: "Management of acute pancreatitis."

A 28-year-old man presents for routine medical evaluation. He reports no specific complaints but requests testing for sexually transmitted infections (STIs). He reports having multiple sexual partners. Testing reveals a positive HIV antibody test, confirmed by Western blot. His CD4 count is 450 cells/mm³ and his viral load is 50,000 copies/mL. Which of the following is the MOST appropriate first-line antiretroviral therapy (ART) regimen to initiate?

- ☐ Two nucleoside reverse transcriptase inhibitors (NRTIs) plus a non-nucleoside reverse transcriptase inhibitor (NNRTI)
- ☐ Two NRTIs plus a protease inhibitor (PI) boosted with ritonavir or cobicistat
- ☒ Two NRTIs plus an integrase strand transfer inhibitor (INSTI)
- ☐ A single-tablet regimen containing an INSTI, an NRTI, and a pharmacokinetic enhancer

Explanation :

First-line therapy for HIV involves combination antiretroviral therapy (ART), typically consisting of two NRTIs (backbone) and a third agent, usually an INSTI due to its potency, favorable side effect profile, and low resistance potential.

Preferred NRTI pairs:

- Tenofovir disoproxil fumarate (TDF) or tenofovir alafenamide (TAF) + emtricitabine (FTC).
- Alternatives include abacavir (ABC) + lamivudine (3TC) (requires HLA-B testing for hypersensitivity risk).

*INSTI is Preferred due to superior efficacy in reducing viral load, minimal drug-drug interactions and better tolerability compared to protease inhibitors (PIs) or NNRTIs.

- UpToDate: "Overview of antiretroviral agents used to treat HIV."

A 45-year-old woman presents with complaints of intermittent bloody stools for the past 6 months. She also reports increasing fatigue and weakness. She denies abdominal pain, weight loss, or changes in bowel habits. Physical examination, including digital rectal examination, is unremarkable. Which of the following is the MOST appropriate next investigation?



A 45-year-old woman presents with complaints of intermittent bloody stools for the past 6 months. She also reports increasing fatigue and weakness. She denies abdominal pain, weight loss, or changes in bowel habits. Physical examination, including digital rectal examination, is unremarkable. Which of the following is the MOST appropriate next investigation?

☐ Flexible sigmoidoscopy

☒ Colonoscopy

☐ Abdominal CT scan

☐ Stool culture

Explanation :

Colonoscopy evaluates the entire colon and rectum, which is necessary to identify potential causes of bleeding such as colorectal cancer, polyps, or angiodysplasia. Intermittent bloody stools: Suggests lower gastrointestinal bleeding, which could be due to polyps, colorectal cancer, or inflammatory bowel disease. Fatigue and weakness may indicate chronic blood loss leading to iron deficiency anemia. Flexible sigmoidoscopy examines only the distal colon and rectum and may miss proximal lesions (e.g., right-sided colon cancer).

- UpToDate: "Evaluation of occult gastrointestinal bleeding."

49

A 55-year-old man presents with persistent dyspepsia despite trying various over-the-counter and prescription medications, including proton pump inhibitors (PPIs) and H₂ receptor antagonists. He reports epigastric burning and bloating, especially after meals. Stool H. pylori antigen test is negative. Which of the following is the MOST appropriate next step in management?

☐ Trial of prokinetic agents

☒ Upper endoscopy with biopsies

☐ Abdominal ultrasound

☐ Empiric treatment for *H. pylori*

Explanation :

50

There is an outbreak of acute watery diarrhea (AWD) in a community. The cumulative incidence is calculated to be 0.1 in one week. Which of the following is the correct interpretation of this finding?

☒ There are 10 new cases per 100 people in one week.

☐ There is 1 new case per 1000 people in one week.

☐ There are 100 new cases per 1000 people in one week.

☐ There is 1 new case per 100 people in one week.

Explanation :

Cumulative Incidence is the proportion of an at-risk population that develops a new disease during a specified period.

• Cumulative Incidence = Number of New Cases/Total Population at Risk

A value of 0.1 implies 10% of the population (or 10 out of 100 individuals) developed the condition within the specified period. A cumulative incidence of 0.1 corresponds to 10 cases per 100 individuals.

- UpToDate: "Measures of disease frequency and association."



51

This question requires specific knowledge of the 1993 Ethiopian Health Policy. I cannot provide the specific details of the policy as it is a complex document with many facets. However, a typical USMLE-style question might ask about the core principles or focus areas of the policy. Which of the following was a key focus of the 1993 Ethiopian Health Policy?

- ☐ Primarily focusing on tertiary care hospitals
- ☒ Emphasis on primary health care and community participation
- ☐ Centralized health system management
- ☐ Exclusive reliance on private healthcare providers

Explanation :

The policy was health-focused and equitable, aimed at addressing the health needs of the population through primary health care (PHC). It emphasized community involvement in planning and implementation to ensure sustainability and effectiveness. Decentralization was a major goal, shifting from a centralized system to one where regional states had autonomy to manage health services. Focus was placed on preventive care, maternal and child health, and combating communicable diseases.

The primary focus of the policy was primary health care, community-based interventions, and preventative health approaches rather than high-cost tertiary care or a privatized system.
- Ethiopian Health Sector Development Program I: Overview of the principles in line with the 1993 policy.

52

A public health team is planning an intervention to reduce the incidence of a specific disease. They are currently in the stage of identifying the most at-risk groups and determining the best methods to reach them. Which of the following BEST describes this stage of the intervention planning process?

- ☐ Setting strategies
- ☒ Situation analysis
- ☐ Implementation
- ☐ Evaluation

Explanation :

53

A 60-year-old man with a history of osteoarthritis presents with black tarry stools (melena) and coffee-ground emesis. He has been taking prednisone for several months for his arthritis. On physical examination, he has epigastric tenderness and guarding. Which of the following is the BEST initial diagnostic investigation?

- ☐ Abdominal ultrasound
- ☒ Upper endoscopy
- ☐ Abdominal CT scan
- ☐ Barium swallow study

Explanation :

Black tarry stools (melena), Coffee-ground emesis Indicates upper gastrointestinal (GI) bleeding.



53

A 60-year-old man with a history of osteoarthritis presents with black tarry stools (melena) and coffee-ground emesis. He has been taking prednisone for several months for his arthritis. On physical examination, he has epigastric tenderness and guarding. Which of the following is the BEST initial diagnostic investigation?

☐ Abdominal ultrasound

☒ Upper endoscopy

☐ Abdominal CT scan

☐ Barium swallow study

Explanation :

Black tarry stools (melena), Coffee-ground emesis indicates upper gastrointestinal (GI) bleeding. Prednisone use is associated with gastric mucosal injury, increasing the risk of peptic ulcer disease (PUD). Upper endoscopy can visualize the source of bleeding, such as peptic ulcers or esophagitis, and facilitate treatment (e.g., coagulation or injection therapy). It is the first-line investigation for patients with signs of upper GI bleeding.

- UpToDate: "Approach to upper gastrointestinal bleeding in adults."

54

A 27-Year-old man has come with complaints of palpitation and exercise intolerance of three months duration. His investigation result shows Hgb level of 5.7mg/dl with MCV of 56fl and iron studies consistent with iron deficiency. The physician following this patient has decided to start him on Ferrous sulfate. What is the most appropriate duration of treatment in this case?

☒ Six-twelve months after correction of Hgb

☐ b. Total of six months

☐ c. Three months after correction of Hgb

☐ d. Total of three months

Explanation :

The main goal of treatment is to correct anemia and replenish iron stores. After hemoglobin levels normalize (typically in 6–8 weeks), iron supplementation should continue for 6–12 months to rebuild iron reserves in the bone marrow and tissues. In iron-deficiency anemia, iron stores in the liver, spleen, and bone marrow are often depleted, and restoring these stores requires several months. Stopping iron supplementation prematurely (e.g., after normalization of hemoglobin) risks recurrence of anemia due to incomplete replenishment of iron.

Harrison's Principles of Internal Medicine: Recommends continuing iron therapy for 6–12 months after correction of hemoglobin.

UpToDate: "Oral iron supplementation is typically continued for 6–12 months after hemoglobin normalization to ensure complete repletion of body iron stores."

55

A group of researchers planned to assess the prevalence of depression and associated factors in an urban community comprising individuals who are able to read and write. A questionnaire containing sensitive questions was prepared in English and translated to local language. What is the most appropriate data collection tool for the above scenario?

☐ Telephone interviewing



55

A group of researchers planned to assess the prevalence of depression and associated factors in an urban community comprising individuals who are able to read and write. A questionnaire containing sensitive questions was prepared in English and translated to local language. What is the most appropriate data collection tool for the above scenario?

☐ Telephone interviewing

☒ Self-administered questionnaire

☐ Observation

☐ Face to face interview

Explanation :

The population is literate (able to read and write), making self-administration feasible. This method ensures anonymity, which is crucial for collecting accurate responses to sensitive questions like those about depression. Participants may respond more honestly without the presence of an interviewer. Self-administered questionnaires are less resource-intensive compared to face-to-face interviews.

56

A 22-year-old boy went to a health center for HIV/AIDS testing. The healthcare provider working at that health facility shared the test result to the boy's father without the knowledge of the client. What is the most likely ethical principle violated in this scenario?

☐ Beneficence

☐ Informed consent

☒ Confidentiality

☐ Fairness and justice

Explanation :

Confidentiality involves protecting personal and sensitive information shared by a patient. It is a cornerstone of medical ethics and is essential for maintaining trust in the patient-provider relationship. The provider's action of sharing the test result without the patient's knowledge or consent constitutes a breach of confidentiality.

HIV/AIDS test results are highly sensitive. Sharing this information without consent violates the patient's right to privacy and can lead to significant psychosocial harm. Ethical guidelines (e.g., WHO and AMA) strictly prohibit disclosure of such results without the patient's permission, except in specific legal or public health contexts.

57

A nine-year-old male child was brought by his parents to a surgical emergency OPD with complaint of abdominal pain and anorexia of four days duration. The pain was initially periumbilical then it involves the whole abdomen. The patient also had high-grade intermittent fever. On physical examination, he was acutely sick looking, pulse rate was 128/min and respiratory rate was 82/min and temperature was 99.6 degree centigrade. The abdomen was tender and rigid. What is the most likely diagnosis?

☐ Simple appendicitis

☐ Mesenteric lymphadenitis

☐ Perforated PUD

